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Multi-Label Emotion Classification for Digital Journaling Applications

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Problem Statement

- Digital journaling market growing rapidly ($6.68B → $15.1B)
- Most apps lack intelligent emotion analysis
- No personalized wellness insights

The Solution: AI-powered journaling with real-time emotion detection

- User writes → BERT analyzes → Detects emotions → Personalized recommendations

System Architecture

React Native App (iOS)

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User writes journal entry

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ML Model: BERT (110M parameters)

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Detects: Joy, Sadness, Anger, Fear, Neutral

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Generates personalized wellness tips

Model Architecture

Input Text (max 48 tokens)

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BERT Encoder (110M parameters, 12 layers)

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Dropout (0.11)

↓

Linear Classifier (768 → 5 emotions)

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Multi-label Predictions

Hyperparameter Optimization

Trial	Val Loss	Learning Rate	Dropout	Batch
1	**0.1925**	**1.15e-05**	**0.11**	**32**
0	0.1988	1.08e-05	0.27	16
2	0.2146	2.91e-05	0.24	32

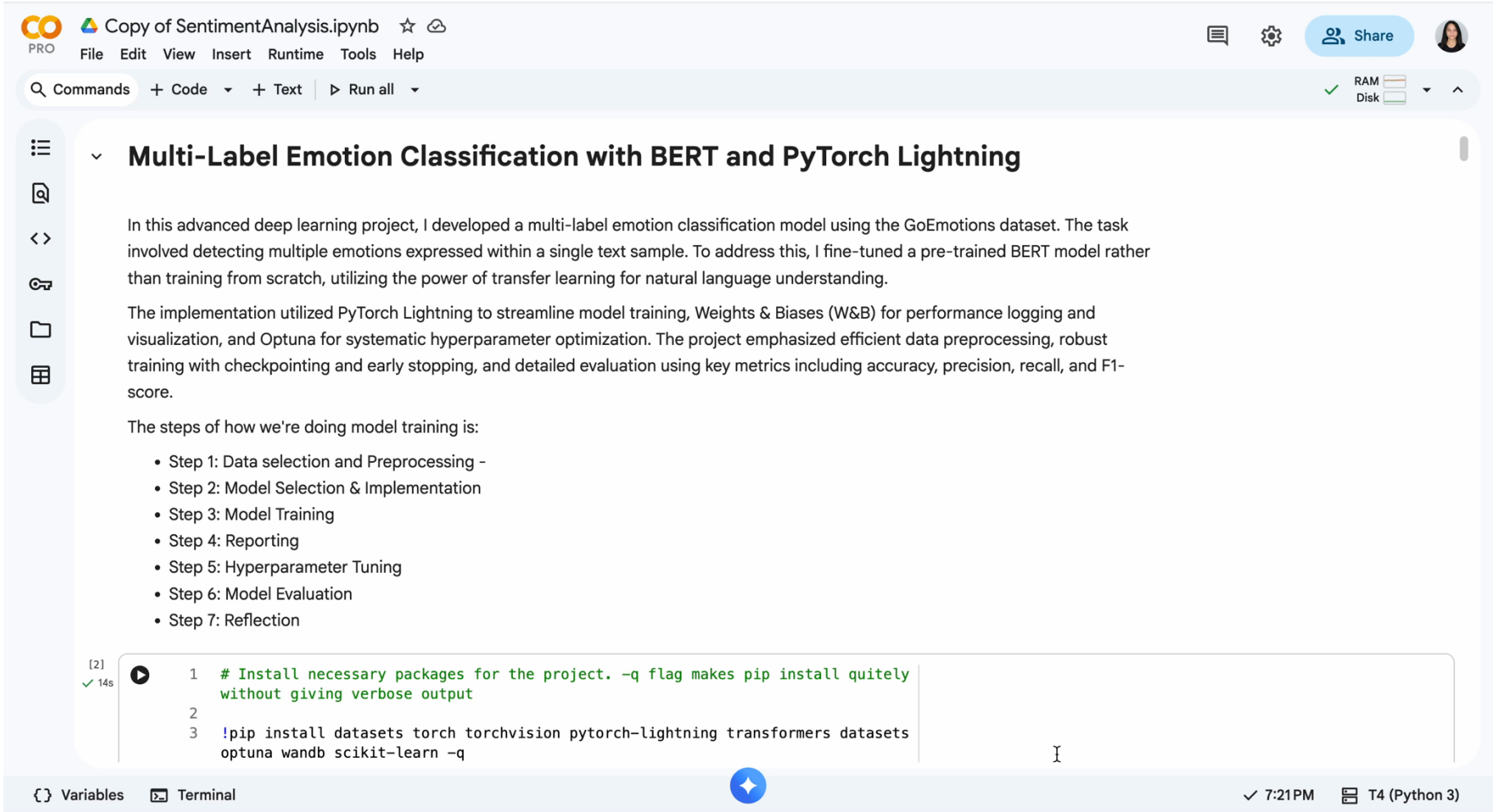
- **Optuna Search Results (3 trials):**
- **Impact:** Baseline 72% → Optimized 88% (+16% improvement)

Training Results

- **Performance Across Datasets:**
- **Test Metrics:** Precision 54.29% | Recall 28.25% | F1 36.47%
- **Analysis:** 27.6% gap indicates overfitting to validation set

Dataset	Accuracy	Loss
Training	94.1%	0.113
Validation	92.65%	0.170
Test	**65.06%**	**0.201**

Model Training Demo



Key Results & Analysis

Strengths:

- Exceeds baseline (20% random → 65% actual)
- High precision (54%) - reliable predictions
- Successfully handles multi-label emotions

Challenges:

- Overfitting (27.6% validation-test gap)
- Low recall (28%) - conservative predictions
- Domain mismatch (Reddit vs journals)

Limitation

Technical Limitations:

1. Overfitting to validation set
2. Low recall on rare emotions
3. Large model size (437MB)
4. Domain adaptation needed

Future Improvements

- Reduce overfitting
- Data augmentation for rare emotions
- Model compression
- Fine-tune on real journal entries
- Expand to 28 emotions
- Personalized user models

Q&A

- Ask me whatever questions you may have

ThankYou